Practice workbook · Merge sort: split, then merge

Showing the split and merge stages of a merge sort on a given list, ascending or descending.

2 question parts: 1 worked-example retry and 1 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-09

1. Question 2 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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1. Question 2 · part c · 5 marks

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Worked-example retry · Attempt part c only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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1. Question 2 · part c · 5 marks

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Worked-example retry · Attempt part c only.

(c) A merge sort algorithm can be used to sort a list into ascending order.

The list 9, 1, 7, 6, 3, 5, 2, 8 needs to be sorted.

Complete the merge sort using the space provided.

(5)

9, 1, 7, 6, 3, 5, 2, 8

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1, 2, 3, 5, 6, 7, 8, 9

(Total for Question 2 = 19 marks)

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1. Question 2 · part c · 5 marks · Working and answer

Working and answer

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2. Question 5 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

5 A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

Figure 5 shows the lists created at the end of the splitting process.

Each list is a single tuna species.

Bigeye	Blackfin	Albacore	Longtail	Bluefin

Figure 5

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

(2)

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2. Question 5 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

Figure 6 shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

Figure 6

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

Line number with error	
Corrected line of pseudocode	

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12

P 7 2 9 3 8 A 0 1 2 1 6

2. Question 5 · part a · 2 marks · Working and answer

Working and answer

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